

CMPE 240

Boolean Algebra

Reference:

Vahid, Appendix A

“plus” / “OR”

“times” / “AND”

A **Boolean Algebra** is a 3-tuple $\{B, +, \cdot\}$, where

- B is a set of at least 2 elements
- $(+)$ and $(\cdot)$ are binary operations (i.e. functions $B \times B \rightarrow B$)

satisfying the following axioms:

A1. Commutative laws: For every $a, b \in B$

I. $a + b = b + a$

II. $a \cdot b = b \cdot a$

A2. Distributive laws: For every $a, b, c \in B$

I. $a + (b \cdot c) = (a + b) \cdot (a + c)$

II. $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

A3. Existence of **identity** elements: The set B has two distinct identity elements, denoted as 0 and 1, such that for every element $a \in B$

I. $a + 0 = 0 + a = a$

additive identity element

II. $a \cdot 1 = 1 \cdot a = a$

multiplicative identity element

A4. Existence of a **complement**: For every element $a \in B$ there exists an element a' such that

I. $a + a' = 1$

II. $a \cdot a' = 0$

the complement of a

Precedence ordering: $\cdot$ **before** $+$

For example:

$$a + (b \cdot c) = a + bc$$

Switching Algebra

$$B = \{ 0, 1 \}$$

AND	0	1
0	0	0
1	0	1

OR	0	1
0	0	1
1	1	1

Theorem 1: The switching algebra is a Boolean algebra.

Proof:

By satisfying the axioms of Boolean algebra:

- B is a set of at least two elements

$$B = \{0, 1\}, 0 \neq 1 \text{ and } |B| \geq 2.$$

- Closure of $(+)$ and $(\cdot)$ over B (functions $B \times B \rightarrow B$).

AND	0	1
0	0	0
1	0	1

OR	0	1
0	0	1
1	1	1

closure

A1. Commutativity of $(+)$ and $(\cdot)$.

AND	0	1
0	0	0
1	0	1

OR	0	1
0	0	1
1	1	1

Symmetric about the main diagonal

A2. Distributivity of $(+)$ and $(\cdot)$. Shown below by perfect induction.

abc	$a + bc$	$(a + b)(a + c)$
000	0	0
001	0	0
010	0	0
011	1	1
100	1	1
101	1	1
110	1	1
111	1	1

abc	$a(b + c)$	$ab + ac$
000	0	0
001	0	0
010	0	0
011	0	0
100	0	0
101	1	1
110	1	1
111	1	1

* Alternative proof of the distributive laws:

Claim: (follow directly from operators table)

- $\text{AND}(0, x) = 0$ $\text{AND}(1, x) = x$
- $\text{OR}(1, x) = 1$ $\text{OR}(0, x) = x$

Consider the distributive law of $(\cdot)$:

$$\text{AND}(a, \text{OR}(b, c)) = \text{OR}(\text{AND}(a, b), \text{AND}(a, c))$$

$$\underline{a = 0} : \underbrace{\text{AND}(0, \text{OR}(b, c))}_{0} = \text{OR}(\underbrace{\text{AND}(0, b)}_{0}, \underbrace{\text{AND}(0, c)}_{0})$$

0

$$\underline{a = 1} : \underbrace{\text{AND}(1, \text{OR}(b, c))}_{\text{OR}(b, c)} = \text{OR}(\underbrace{\text{AND}(1, b)}_{b}, \underbrace{\text{AND}(1, c)}_{c})$$

$\text{OR}(b, c)$



Consider the distributive law of (+):

$$\text{OR}(a, \text{AND}(b, c)) = \text{AND}(\text{OR}(a, b), \text{OR}(a, c))$$

$$\underline{a = 0} : \underbrace{\text{OR}(0, \text{AND}(b, c))}_{\text{AND}(b, c)} = \text{AND}(\underbrace{\text{OR}(0, b)}_b, \underbrace{\text{OR}(0, c)}_c)$$

$\underbrace{\hspace{10em}}_{\text{AND}(b, c)}$

$$\underline{a = 1} : \underbrace{\text{OR}(1, \text{AND}(b, c))}_1 = \text{AND}(\underbrace{\text{OR}(1, b)}_1, \underbrace{\text{OR}(1, c)}_1)$$

$\underbrace{\hspace{10em}}_1$



Why have we done that?

For complex expressions truth tables are not an option.

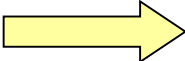
A3. Existence of additive and multiplicative identity element.

$$0 + 1 = 1 + 0 = 1 \quad \longrightarrow \quad 0 - \text{additive identity}$$

$$0 \cdot 1 = 1 \cdot 0 = 0 \quad \longrightarrow \quad 1 - \text{multiplicative identity}$$

A4. Existence of the complement.

a	a'	$a + a'$	$a \cdot a'$
1	0	1	0
0	1	1	0

All axioms are satisfied  Switching algebra is Boolean algebra.



Theorems in Boolean Algebra

Theorem 2:

Every element in B has a **unique** complement.

Proof:

Let $a \in B$. Assume that a_1' and a_2' are both complements of a , (i.e. $a_i' + a = 1$ & $a_i' \cdot a = 0$), we show that $a_1' = a_2'$.

Identity

$$a_1' = a_1' \cdot 1$$

a_2' is the complement of a

$$= a_1' \cdot (a + a_2')$$

distributivity

$$= a_1' \cdot a + a_1' \cdot a_2'$$

commutativity

$$= a \cdot a_1' + a_1' \cdot a_2'$$

a_1' is the complement of a

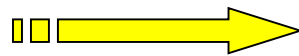
$$= 0 + a_1' \cdot a_2'$$

Identity

$$= a_1' \cdot a_2'$$

We swap a_1' and a_2' to obtain,

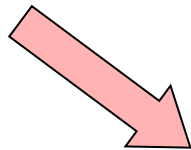
$$\begin{aligned} a_2' &= a_2' \cdot a_1' \\ &= a_1' \cdot a_2' \end{aligned}$$



$$a_1' = a_2'$$



★ Complement uniqueness



, can be considered as a unary operation

$B \rightarrow B$ called **complementation**

Boolean Algebra vs. Numbers Operations

- Is the following theorem correct?
 - if there exists x such that: $x + a = x + b$
then $a = b$ (elimination of x from both sides).
- Consider the switching algebra, and choose $x = 1$, $a = 0$, $b = 1$.
- For numbers, there is a minus operation, which eliminates x .

Boolean expression - Recursive definition:

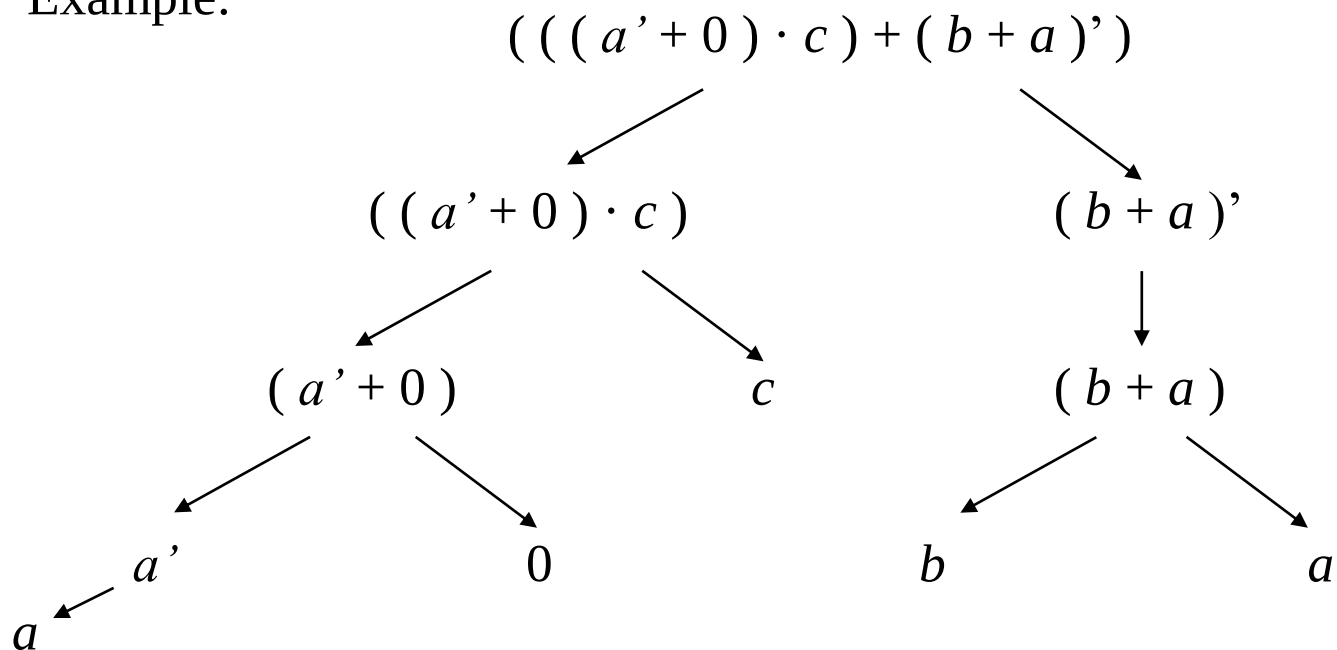
base: $0, 1, a \in B$ – expressions.

recursion step: Let E_1 and E_2 be Boolean expressions.

Then,

$$\left. \begin{array}{l} E_1' \\ (E_1 + E_2) \\ (E_1 \cdot E_2) \end{array} \right\} \text{expressions}$$

Example:



Dual transformation - Recursive definition:

Dual: expressions $\rightarrow$ expressions

base: $0 \rightarrow 1$

$1 \rightarrow 0$

$a \rightarrow a, a \in B$

recursion step: Let E_1 and E_2 be Boolean expressions.

Then,

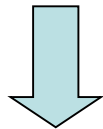
$E_1' \rightarrow [\text{dual}(E_1)]'$

$(E_1 + E_2) \rightarrow [\text{dual}(E_1) \cdot \text{dual}(E_2)]$

$(E_1 \cdot E_2) \rightarrow [\text{dual}(E_1) + \text{dual}(E_2)]$

Example:

$$((a + b) + (a' \cdot b')) \cdot 1$$



$$((a \cdot b) \cdot (a' + b')) + 0$$



The axioms of Boolean algebra are in dual pairs.

A1. Commutative laws: For every $a, b \in B$

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I. $a + (b \cdot c) = (a + b) \cdot (a + c)$

II. $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

A3. Existence of identity elements: The set B has two distinct identity elements, denoted as 0 and 1, such that for every element $a \in B$

I. $a + 0 = 0 + a = a$

II. $a \cdot 1 = 1 \cdot a = a$

A4. Existence of a complement: For every element $a \in B$ there exists an element a' such that

I. $a + a' = 1$

II. $a \cdot a' = 0$

Theorem 3: (null elements)

For every $a \in B$:

1. $a + 1 = 1$

2. $a \cdot 0 = 0$

Proof:

(1)

Identity

$$a + 1 = 1 \cdot (a + 1)$$

a' is the complement of a

$$= (a + a') \cdot (a + 1)$$

distributivity

$$= a + (a' \cdot 1)$$

Identity

$$= a + a'$$

a' is the complement of a

$$= 1$$

(2) we can do the same way:

Identity	$a \cdot 0 = 0 + (a \cdot 0)$
a' is the complement of a	$= (a \cdot a') + (a \cdot 0)$
distributivity	$= a \cdot (a' + 0)$
Identity	$= a \cdot a'$
a' is the complement of a	$= 0$



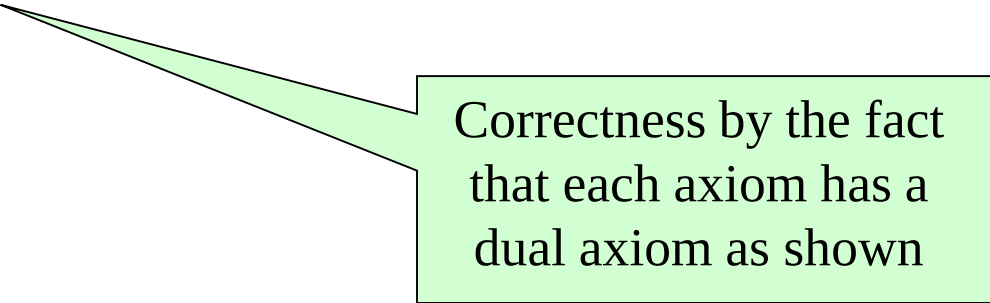
Note that:

- $a \cdot 0, 0$ are the dual of $a + 1, 1$ respectively.
- The proof of (2) follows the same steps exactly as the proof of (1) with the same arguments, but applying the dual axiom in each step.

Theorem 4: Principle of Duality

Every proof that $E_1 = E_2$

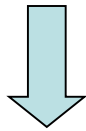
Can be transformed into a “dual” proof that $\text{dual}(E_1) = \text{dual}(E_2)$



Correctness by the fact
that each axiom has a
dual axiom as shown

For example:

$$(a + b) + a' \cdot b' = 1$$



$$(a \cdot b) \cdot (a' + b') = 0$$



Every theorem proves its dual for “free”
see Theorem 3

Theorem 5:

The complement of the element 1 is 0, and vice versa:

$$1. \quad 0' = 1$$

$$2. \quad 1' = 0$$

Proof:

By Theorem 3, we have for every $a \in B$: $a + 1 = 1$, $a \cdot 0 = 0$,

$$0 + 1 = 1 \text{ and } 0 \cdot 1 = 0 \quad (A)$$

From complement Axiom

$$0 + 0' = 1 \text{ and } 0 \cdot 0' = 0 \text{ then } 0' = 1$$

Both 1 and $0'$ are complements of 0. By the uniqueness of the complement from Theorem 2 and the complement axiom, the Theorem follows. ■

Remember complement axiom:

$$\text{I. } a + a' = 1$$

$$\text{II. } a \cdot a' = 0$$

Theorem 6: Idempotent Law

For every $a \in B$

1. $a + a = a$

2. $a \cdot a = a$

Proof:

(1)

Identity	$a + a = (a + a) \cdot 1$
a' is the complement of a	$= (a + a) \cdot (a + a')$
distributivity	$= (a + (a \cdot a'))$
a' is the complement of a	$= a + 0$
Identity	$= a$

(2) duality.



Theorem 7: Involution Law

For every $a \in B$

$$(a')' = a$$

Proof:

From the definition of complement,

$$\text{I. } a + a' = 1$$

$$\text{II. } a \cdot a' = 0$$

We see that a' has two complements $(a')'$ and a .

But by the uniqueness of complement from Theorem 2, Theorem 7 is proved.



Theorem 8: Absorption Law

For every pair of elements $a, b \in B$,

$$1. \quad a + a \cdot b = a$$

$$2. \quad a \cdot (a + b) = a$$

Proof:

(1)

$$\begin{aligned} a + a \cdot b &= a \cdot 1 + a \cdot b \text{ (identity)} \\ &= a \cdot (1 + b) \text{ (distributivity)} \\ &= a \cdot (b + 1) \text{ (commutativity)} \\ &= a \cdot 1 \text{ (Theorem 3, null elements)} \\ &= a \text{ (identity)} \end{aligned}$$

(2) duality.



Theorem 9:

For every pair of elements $a, b \in B$,

1. $a + a' \cdot b = a + b$
2. $a \cdot (a' + b) = a \cdot b$

Proof:

(1)

distributivity

$$a + a'b = (a + a')(a + b)$$

a' is the complement of a

$$= 1(a + b)$$

Identity

$$= a + b$$

(2) duality. ■

Theorem 10:

In a Boolean algebra, each of the binary operations $(+)$ and $(\cdot)$ is associative. That is, for every $a, b, c \in B$,

$$1. \quad a + (b + c) = (a + b) + c$$

$$2. \quad a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

Proof: **home assignment** (hint: prove that both sides in (1) equal $[(a + b) + c] \cdot [a + (b + c)]$.)

Theorem 11: DeMorgan's Law

For every pair of elements $a, b \in B$,

$$1. \quad (a + b)' = a' \cdot b'$$

$$2. \quad (a \cdot b)' = a' + b'$$

Proof: home assignment.

Theorem 12: Generalized DeMorgan's Law

Let $\{a, b, \dots, c, d\}$ be a set of elements in a Boolean algebra.

Then, the following identities hold:

$$1. (a + b + \dots + c + d)' = a' b' \dots c' d'$$

$$2. (a \cdot b \cdot \dots \cdot c \cdot d)' = a' + b' + \dots + c' + d'$$

Proof: By **induction**.

Induction basis: follows from DeMorgan's Law

$$(a + b)' = a' \cdot b'.$$

Induction hypothesis: DeMorgan's law is true for n elements.

Induction step: show that it is true for $n+1$ elements.

Let $a, b, \dots, c$ be the n elements, and d be the $(n+1)^{st}$ element.

$$(a + b + \dots + c + d)' = [(a + b + \dots + c) + d]'$$

Associativity

$$= (a + b + \dots + c)' d'$$

DeMorgan's Law

$$= a'b' \dots c'd'$$

Induction assumption $(a + b + \dots + c)' = a'b' \dots c'$





The symbols $a, b, c, \dots$ appearing in theorems and axioms are **generic variables**

Can be substituted by complemented variables or expressions (formulas)

For example:

$$(a + b)' = a' b'$$

DeMorgan's Law

$$a \leftarrow a'$$

$$b \leftarrow b'$$

$$(a' + b')' = ab$$

$$a \leftarrow (a + b)$$

$$b \leftarrow c'$$

$$[(a + b) + c']' = (a + b)' c$$

etc.

Other Examples of Boolean Algebras

Algebra of Sets

Consider a set S .

B = all the subsets of S (denoted by $P(S)$).

“plus” $\rightarrow$ set-union $\cup$

“times” $\rightarrow$ set-intersection $\cap$

$$M = (P(S), \cup, \cap)$$

Additive identity element – empty set $\emptyset$

Multiplicative identity element – the set S .

$P(S)$ has $2^{|S|}$ elements, where $|S|$ is the number of elements of S

Algebra of Logic (Propositional Calculus)

Elements of B are T and F (true and false).

“plus” $\rightarrow$ Logical OR $\vee$

“times” $\rightarrow$ Logical AND $\wedge$

$$M = (\{T, F\}, \vee, \wedge)$$

Additive identity element – F

Multiplicative identity element – T